Year 11 Set 4 · Foundation · Lesson 4

Area & perimeter, including circles

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Today I will learn to...

Goals

1. Find the area of triangles, parallelograms and trapeziums

2. Find the area and circumference of circles

Do now: retrieval practice

4 mins

1 Solve $3x - 2 = 13$

$x = 5$

2 Find 15% of 60

9

3 Work out 4.5×2

9

4 Round 28.274 to 1 dp

28.3

Objective 1

Find the area of triangles, parallelograms and trapeziums

Key idea

Learn

!

Triangle = ½ × b × h. Parallelogram = b × h. Trapezium = ½(a + b)h. Height is always perpendicular to the base.

Objective 1: Find the area of triangles, parallelograms and trapeziums

I do: example 1

I do

Triangle: base 8 cm, height 5 cm

1. Area of a triangle = ½ × base × height
2. = ½ × 8 × 5
3. = ½ × 40
4. = 20 cm²

I do: example 2

I do

Parallelogram: base 9 cm, height 4 cm

1. Area of a parallelogram = base × height
2. Use the perpendicular height, not the slant side
3. $= 9 \times 4$
4. $= 36\ \text{cm}^2$

I do: example 3

I do

Trapezium: parallel sides 6 cm and 10 cm, height 5 cm

1. Area of a trapezium = ½ × (a + b) × h
2. Add the parallel sides: 6 + 10 = 16
3. ½ × 16 × 5 = 8 × 5
4. = 40 cm²

Mini whiteboards

We do

1. Triangle: base 12, height 7

Answer: 42

2. Trapezium: sides 3 and 7, height 4

Answer: 20

3. Parallelogram: base 9, height 4

Answer: 36

4. Triangle: base 10, height 3

Answer: 15

You do: choose your challenge

You do

Bronze

Find the area of a rectangle 7 cm by 6 cm.

Silver

Find the area of a triangle with base 9 cm and height 6 cm.

Gold

A trapezium has parallel sides 8 cm and 12 cm and height 5 cm. Find its area.

Platinum

A 12 m by 8 m garden has a triangular flower bed with base 6 m and height 4 m. Turf costs £5 per m^2 for the rest. Find the cost.

You do: answers

Check

Bronze

$42\ cm^2$

Silver

$27\ cm^2$

Gold

$\frac{1}{2} \times 20 \times 5 = 50\ cm^2$

Platinum

$96 - 12 = 84\ m^2$, $84 \times 5 = £420$

If you finish early

Extra

Bronze

Find the area of a triangle with base 8 cm and height 5 cm.

20 cm²

Silver

Find the area of a trapezium with parallel sides 6 cm and 10 cm and height 4 cm.

32 cm²

Gold

A triangle has area 30 cm² and base 12 cm. Find its height.

5 cm

Platinum

A trapezium has area 54 cm², height 6 cm and one parallel side 7 cm. Find the other parallel side.

(7 + b) × 3 = 54, b = 11 cm

Exam question 1

2 marks

A trapezium has parallel sides 5 cm and 11 cm and height 6 cm. Work out its area.

Space for working

Mark scheme: ½ × 16 × 6 = 48 cm²

Exam question 2

2 marks

An L-shape is made from a 10 cm × 4 cm rectangle and a 3 cm × 5 cm rectangle. Work out its area.

Space for working

Mark scheme: 40 + 15 = 55 cm²

Objective 1 achieved

Find the area of triangles, parallelograms and trapeziums



Objective 2

Find the area and circumference of circles

✓ Achieved: Find the area of triangles, parallelograms and trapeziums

Key idea

Learn

!

Area = πr^2. Circumference = πd (or $2\pi r$). Check whether you were given the radius or the diameter.

Objective 2: Find the area and circumference of circles

I do: example 1

I do

Area of a circle with radius 5 cm

1. Area = $\pi \times r^2$
2. $r^2 = 5 \times 5 = 25$
3. Area = $\pi \times 25 = 78.539\ldots$
4. = 78.5 cm^2 (1 dp)

I do: example 2

I do

Circumference of a circle with diameter 12 cm

1. Circumference = $\pi \times d$
2. $d = 12$, so $C = \pi \times 12$
3. $= 37.699\ldots$
4. $= 37.7$ cm (1 dp)

I do: example 3

I do

Area of a semicircle with radius 4 cm

1. Whole circle area = $\pi \times r^2$
2. $\pi \times 4^2 = 16\pi = 50.265\ldots$
3. A semicircle is half: $50.265\ldots \div 2$
4. $= 25.1\ \text{cm}^2$ (1 dp)

Mini whiteboards

We do

1. Circumference, radius 3 (1 dp)

Answer: 18.8

2. Area, radius 10 (1 dp)

Answer: 314.2

3. Area, diameter 6 (1 dp)

Answer: 28.3

4. Circumference, diameter 20 (1 dp)

Answer: 62.8

You do: choose your challenge

You do

Bronze

Area of a circle with radius 4 cm (1 dp)

Silver

Circumference of a circle with diameter 9 cm (1 dp)

Gold

A round table top has radius 0.6 m. Find its area (2 dp).

Platinum

A circle of diameter 10 cm is cut from a square of side 10 cm. Find the area left (1 dp).

You do: answers

Check

Bronze	$50.3\ cm^2$
Silver	$28.3\ cm$
Gold	$1.13\ m^2$
Platinum	$100 - 78.54 = 21.5\ cm^2$

If you finish early

Extra

Bronze

Find the circumference of a circle with radius 5 cm (1 dp).

31.4 cm

Silver

Find the area of a circle with diameter 14 cm (1 dp).

153.9 cm²

Gold

A circle has circumference 50 cm. Find its radius (1 dp).

50 ÷ 2π = 8.0 cm

Platinum

A semicircle has diameter 12 cm. Find its area and its perimeter (1 dp).

Area 56.5 cm², perimeter 6π + 12 = 30.8 cm

Exam question 1

2 marks

A circle has diameter 14 cm. Work out its area to 1 dp.

Space for working

Mark scheme: r = 7, π × 49 = 153.9 cm²

Exam question 2

2 marks

A circular pond has radius 2.5 m. Work out its circumference to 1 dp.

Space for working

Mark scheme: $2 \times \pi \times 2.5 = 15.7$ m

Objective 2 achieved

Find the area and circumference of circles



Real life

Apply

A 30 cm pizza costs £9 and a 20 cm pizza costs £5. Which gives more pizza per pound?

Large: 706.9 cm² ÷ 9 = 78.5 cm² per £. Small: 314.2 ÷ 5 = 62.8 cm² per £. The large is better value.

Extension: stretch yourself

Stretch

1. **Find the perimeter of a quarter circle with radius 6 cm (1 dp).**

Arc = ¼ × 12π = 9.42; + 6 + 6 = 21.4 cm

2. **A circle has area 50 cm². Find its radius (2 dp).**

r = √(50 ÷ π) = 3.99 cm

Exit ticket: exam questions

Exit

1. Find the area of a triangle with base 14 cm and height 5 cm. **(2 marks)**

2. Find the area of a circle with radius 6 cm. Give your answer to 1 dp. **(2 marks)**

3. Find the perimeter of a semicircle with diameter 8 cm. Give your answer to 1 dp. **(3 marks)**

Exit ticket: mark scheme

Mark it

Q1 35 cm² [2]

Q2 113.1 cm² [2]

Q3 Arc = ½ × π × 8 = 12.57. 12.57 + 8 = 20.6 cm [3]

Total: _____ / 7

Today I achieved

- ✔ **Find the area of triangles, parallelograms and trapeziums**
- ✔ **Find the area and circumference of circles**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Find the area of a parallelogram with base 11 cm and height 6 cm. [2]

2. A trapezium has parallel sides 4 cm and 10 cm and height 7 cm. Find its area. [2]

3. Find the circumference of a circle with radius 7.5 cm (1 dp). [2]

4. A wheel has diameter 60 cm. How far does it travel in 100 turns? Give your answer in metres to 1 dp. [3]

5. Find the area of a quarter circle with radius 10 cm (1 dp). [2]

Extra practice: Maths Genie → Edexcel question bank → Area and Circumference of Circles